Name:_____ Student number:_____

MID-TERM BIOL 300: March 2013 (Irwin)

Instructions:
1. Make sure that your copy of the test includes 6 pages, including this one.
2. Answers can be in ink or pencil. If you want the possibility of re-marking, use ink.
4. Other than a non-programmable calculator, no electronic devices are permitted. Please turn off your cell phone and leave it in your bag.
3. Students who engage in dishonest practices will be subject to disciplinary action.
4. By taking this test and putting your name above, you are declaring that your answers on this test are all your own work.
5. Students may not speak or in any other way communicate with other students while in the examination room.
6. Students may not expose their written paper to other students. The excuse of accidental exposure, forgetfulness, or ignorance will not be accepted.

Some questions have a box for the final answer. Please put the final answer in this box, and show your work in the other space provided (use the back of the page if necessary).

I have read and fully understand these instructions.

Student signature _____

Mark allocation:

Question	Marks possible	Your mark
1.	18	
2.	18	
3.	24	
4.	20	
5.	20	
Total	**100**	

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1. Answer each of the following (*3 points each*):

a. Describe the shape of the Z-distribution. What is its mean and standard deviation?

Standard normal distribution. Symmetrical, bell-shaped.

Mean = 0

SD = 1

b. How does a t-distribution differ from a Z-distribution?

Very similar, but less in the middle and more in the tails.

c. What is a histogram, and for what kind of data would you use it?

Used with numerical variables, it is a graph showing the number of individuals within intervals (of equal length spaced along the horizontal axis).

d. Imagine you have a discrete numerical variable. How do you determine the mode?

It is simply the most common value for the variable.

e. What is meant by the "Power" of a test?

The probability of rejecting a false null.

f. Describe the characteristics of a good null hypothesis.

Very specific, and boring.

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2. You plan to conduct a statistical test using $\alpha = 0.05$.

a. Assume the null hypothesis is true. What is the probability of rejecting the null hypothesis? (*3 pts*)

0.05

b. Assume the null hypothesis is false. What is the probability of committing a Type 1 error? (*3 pts*)

Zero. (If the null is false, there is no way to commit a Type 1 error. So the probability is zero.)

c. What is a Type 2 (or Type II) error? (*3 pts*)

Failing to reject a false null hypothesis.

d. You get a P-value of 0.11. What do you conclude? (*3 pts*)

You cannot reject the null hypothesis.

(Note that this answer should get zero credit: "We accept the null hypothesis")

e. You conduct three independent statistical tests, each using $\alpha = 0.05$. Assume each null hypothesis is true. What is the probability that exactly one (and only one) Type 1 error will be made? (*6 pts*)

(3 choose 1) $(0.05)^1 (0.95)^2 = 0.135$

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3. Delmore et al. (2012) used small tracking devices to study the migratory routes of Swainson's Thrushes from their breeding grounds in British Columbia to their wintering grounds in Central and South America. In a sample of 4 of these birds from central BC, the flight speed (in km per day of flight) during fall migration to South America was as follows:

260 550 460 570

Using this sample, calculate estimates of the following summary statistics (remember to include a few decimal places in your calculations, to avoid rounding error):

a. Mean (*3 pts*)

2a. 460

b. Median (*3 pts*)

2b. 505

c. Range (*3 pts*)

2c. 310 (or 260-570)

d. Standard deviation (*4 pts*)

2d. 141.66

e. Standard error of the mean (*4 pts*)

141.66/sqrt(4)

2e. 70.83

f. 95% confidence interval for the population mean, assuming the flight speeds in the population are normally distributed (*7 pts*)

$t_{0.05(2),3} = 3.18$

460 +- SE* $t_{0.05(2),3}$ = 460 +- (70.83)(3.18) = 460 +- 225.2

2f. 460 ± 225 (or 235 to 685)

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4. Streaked Horned Larks are declining in numbers both in Canada and in the state of Washington. In an effort to increase the reproductive rate of this species, Pearson et al. (2012) placed wire cages around nests, in the hopes that this would keep predators away from nests. The researchers then compared the rate of predation on 33 nests with cages (4 were predated, and 29 were not predated) with the rate of predation on 31 nests without cages (15 were predated, and 16 were not predated). Assume that the nests studied were a random sample of all nests, and that nests were randomly assigned to treatment groups (cage or no-cage).

Conduct a statistical test to determine whether you can conclude that predation rate depends on whether nests had cages around them. Clearly state your null and alternative hypotheses and the name of the test you are using. Assume $\alpha = 0.05$. Show your work, and interpret your results in words. If you can, give the *P*-value as precisely as possible from the statistical tables. (*20 pts*)

H_0: Nests with and without cages have an equal probability of being predated.
H_A: Nests with cages do NOT have an equal probability of being predated as nests without cages.

χ^2 Contingency Test

Observed

	cage	No cage	Total
Predated	4	15	19
Not predated	29	16	45
Total	33	31	64

Expected

	cage	No cage	Total
Predated	9.80	9.20	19
Not predated	23.20	21.80	45
Total	33	31	64

χ^2 = sum $((obs-exp)^2/exp) = 10.07$

df = 1
$\chi^2_{crit} = \chi^2_{1,0.05} = 3.84$

Since $\chi^2_{data} > \chi^2_{crit}$, we conclude $P < 0.05$, and reject the null and conclude that nest predation depends on presence of cages. (In fact, we can also conclude that $0.005 < P < 0.001$, by examining the χ^2 table more closely).

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5. What statistical test would you use in each of the following situations? Assume random sampling and normal population distributions of numerical variables. Give the full name of the test, and if appropriate state the frequency distribution used for your null model. (For this question, you do NOT need to do the tests.)

a. You have two categorical variables, and you want to test the null hypothesis that the two variables are independent in the population of interest. Assume that you have quite a large sample size. (*4 pts*)

5a. χ^2 Contingency Test

b. Soccer team A beats soccer team B by a score of 4 to 1. You want to test the null hypothesis that the two teams are in fact equal in ability and have equal chances of scoring each goal. (*4 pts*)

5b. Binomial Test

c. You measure the trunk lengths of 34 adult female elephants, and then want to use the data to test the null hypothesis that the mean trunk length of the population of female elephants is 1.5 meters. (*4 pts*)

5c. (One-sample) *t*-Test

d. You are studying a population of butterflies, and you want to know whether survival depends on sex. You track 15 individuals, and determine for each whether it is male or female, and whether it lives or dies. You want to then test the null hypothesis that the survival rate is the same in both sexes in the population of butterflies. Note that your total sample size is only 15 individuals. (*4 pts*)

5d. Fisher's Exact Test

e. Killer whales can prey on seals or fish. You watch 165 individual killer whales, and for each individual you watch 3 different predation events, recording each time whether a seal or fish is eaten. You then have ~~five~~ four categories (0 seals eaten, 1 seal eaten, 2 seals eaten, 3 seals eaten), with the number of whales in each. You want to test the null hypothesis that each predation event is independent of others, such that the probabilities of eating a seal or fish are always constant. (The alternative possibilities are much more interesting: that some individuals tend to focus on eating one prey species, or alternatively that individuals tend to eat a balanced diet.) (*4 pts*)

5e. χ^2 Goodness-of-Fit Test, using Binomial Dist.